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Course: ITD103 - Modern Query Languages

Lab 2 Part A: Schema Design

1. E-commerce Product Catalog

- **Design Approach:** Embedding for variants, reviews, and categories; Referencing for customer orders.
- **Justification:** Embedding variants and reviews directly inside the product document optimizes read performance, as this information is almost always retrieved together when a user views a product page. However, for orders, a referencing approach is preferred (linking a customer and products to an order ID) to prevent unbounded document growth and maintain data normalization across the catalog.
- **Schema Structure (JSON Representation):**

JSON

```
{
  "_id": "ObjectId",
  "name": "Smartphone",
  "category": "Electronics",
  "price": 699.99,
  "variants": [
    { "color": "Black", "stock": 50 },
    { "color": "White", "stock": 30 }
  ],
  "reviews": [
    {
      "user": "Alice",
      "rating": 5,
      "comment": "Great battery life!"
    }
  ]
}
```

2. Blog System

- **Design Approach:** Embedding for tags and hierarchical comments; Referencing for author profiles.
- **Justification:** Tags and comments are strongly coupled to the specific blog post and are loaded simultaneously when a post is opened, making an embedded array highly efficient. To address the hierarchical comments structure requirement, comments are embedded with nested replies. The author is referenced by an ID to avoid duplicating author profile data (like bio and social links) across every post they write.
- **Schema Structure (JSON Representation):**

JSON

```
{
  "_id": "ObjectId",
  "title": "Understanding Document Databases",
  "author_id": "ObjectId(Reference to Authors Collection)",
  "content": "Full blog post text goes here...",
  "tags": ["NoSQL", "MongoDB", "Database Design"],
  "comments": [
    {
      "comment_id": 1,
      "user": "Bob",
      "text": "Very helpful guide!",
      "replies": [
        {
          "user": "Alice",
          "text": "I agree, especially the part about embedding."
        }
      ]
    }
  ]
}
```